

機械手臂 逆向運動學

Manipulator Inverse Kinematics

林沛群

國立台灣大學

機械工程學系



引言

□ 手臂順向運動學 Forward kinematics (FK)

給予 θ_i (可計算出 ${}^{i-1}_iT$) , 求得 $\{H\}$ 或 wP

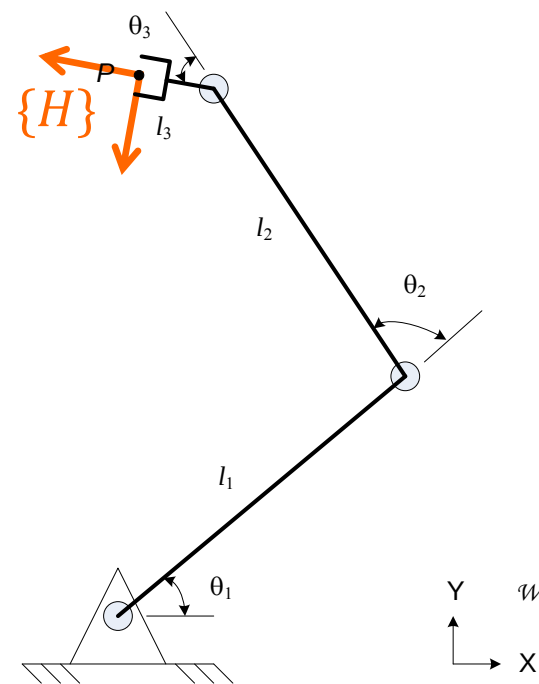
$${}^w_HT = f(\theta_1, \dots, \theta_i, \dots, \theta_n)$$

$${}^wP = {}^w_HT {}^HP$$

□ 手臂逆向運動學 Inverse kinematics (IK)

給予 $\{H\}$ 或 wP , 求得 θ_i

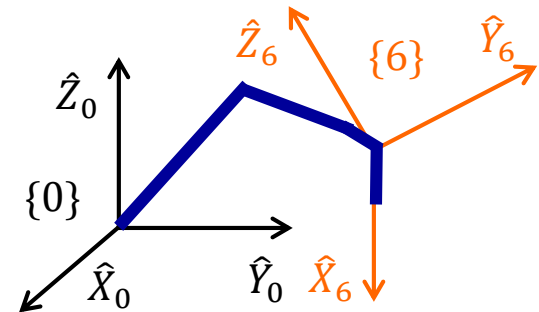
$$[\theta_1, \dots, \theta_i, \dots, \theta_n] = f^{-1}({}^w_HT)$$



求解概念 -1

□ 假設手臂有6 DOFs

- ◆ 6 個未知的joint angles (θ_i 或 d_i , $i = 1, \dots, 6$)



□ 在 ${}^w_H T$ 中擷取出含未知數的 ${}^0_6 T$ ，16個數字

$${}^0_6 T = \begin{bmatrix} {}^0_6 R_{3 \times 3} & {}^0 P_{6 \text{ org } 3 \times 1} \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4 \times 4} = \begin{bmatrix} | & | & | & | \\ {}^0 \hat{X}_6 & {}^0 \hat{Y}_6 & {}^0 \hat{Z}_6 & {}^0 P_{6 \text{ org }} \\ | & | & | & | \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

□ 求解

- ◆ 12個nonlinear transcendental equations方程式
- ◆ 6個未知數，6個限制條件

求解概念 -2

□ Reachable workspace

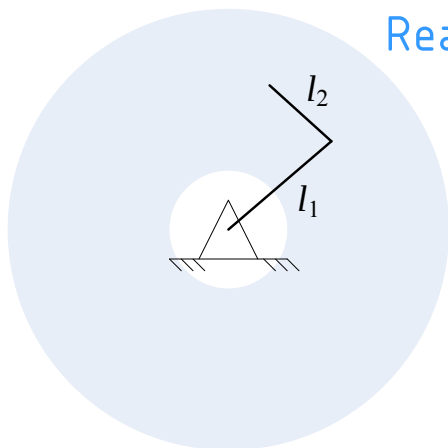
- ◆ 手臂可以用一種以上的姿態到達的位置

□ Dexterous workspace

- ◆ 手臂可以用任何的姿態到達的位置

□ Ex: A RR manipulator

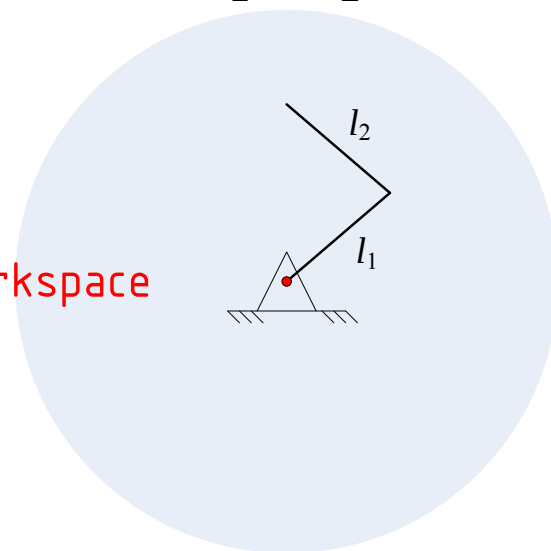
If $l_1 > l_2$



Reachable workspace

Dexterous workspace

If $l_1 = l_2$



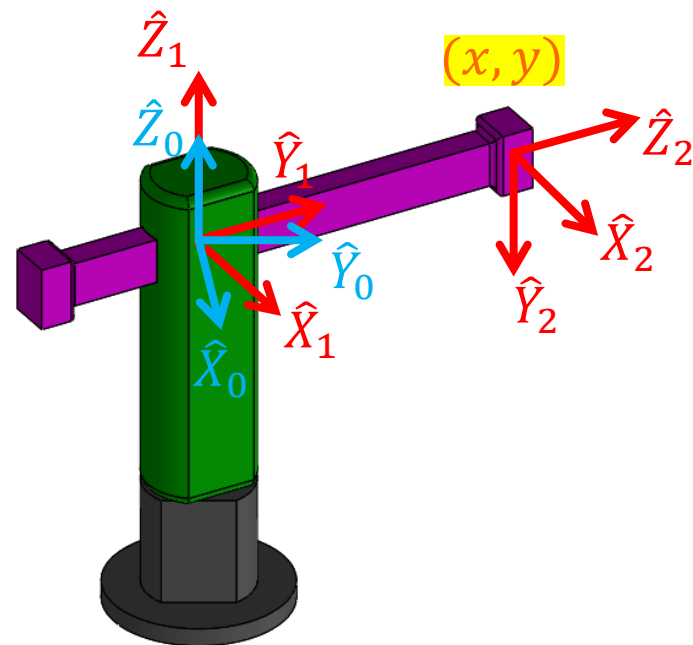
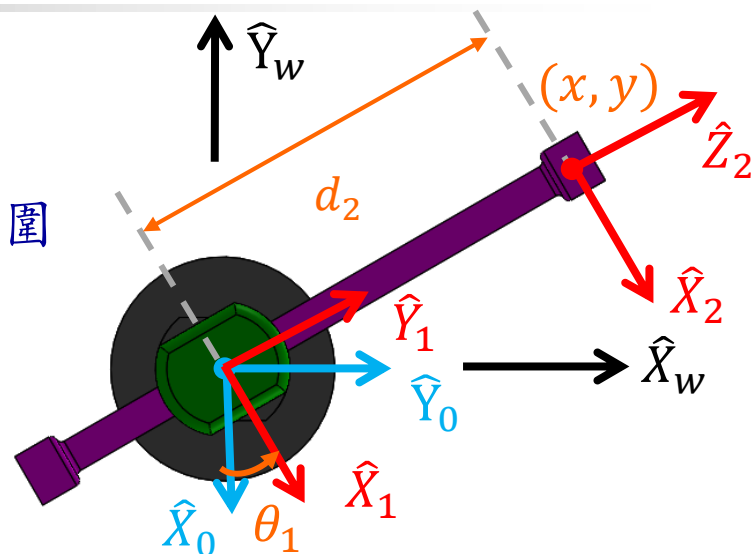
求解概念 -3

□ Subspace

- ◆ 手臂在定義頭尾的 T 所能達到的變動範圍

□ Ex: A RP manipulator

- ◆ 2-DOF, Variables: (x, y)



$${}^w_2T = \begin{bmatrix} \frac{y}{\sqrt{x^2 + y^2}} & 0 & \frac{x}{\sqrt{x^2 + y^2}} & x \\ \frac{-x}{\sqrt{x^2 + y^2}} & 0 & \frac{y}{\sqrt{x^2 + y^2}} & y \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

${}^0\hat{Z}_2$ ${}^0P_{2\text{ ORG}}$

机械臂末端
(x,y)相对
于基点的位
姿

多重解 -1

□ 解的數目

- ◆ 由於是nonlinear transcendental equations，6未知數6方程式不代表具有唯一解
- ◆ 是由joint數目和link參數所決定

Ex: A RRRRRR manipulator

a_i	解的數目
$a_1 = a_3 = a_5 = 0$	≤ 4
$a_3 = a_5 = 0$	≤ 8
$a_3 = 0$	≤ 16
$All\ a_i \neq 0$	≤ 16

多重解 -2

□ Ex: PUMA (6 rotational joints)

- ◆ 針對特定工作點，8組解

- ◆ 前3軸具有4種姿態

如右圖所示

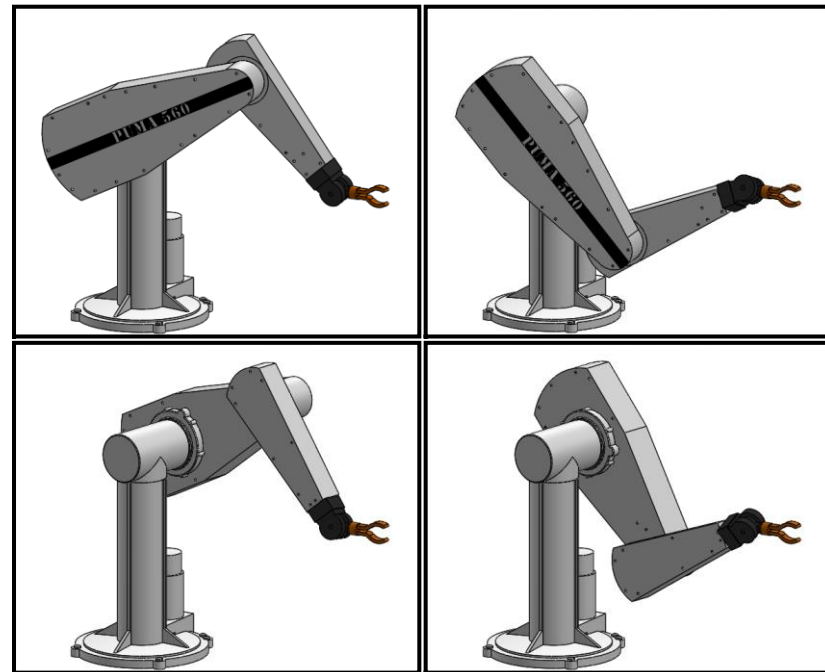
- ◆ 每一個姿態中，具有2組手腕轉動姿態

$$\theta'_4 = \theta_4 + 180^\circ$$

$$\theta'_5 = -\theta_5$$

$$\theta'_6 = \theta_6 + 180^\circ$$

- ◆ 若手臂本身有幾何限制，並非每一種解都可以運作



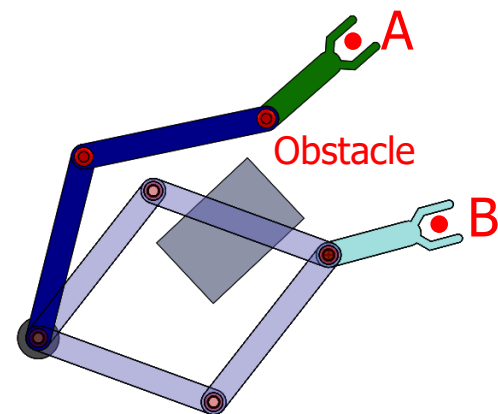
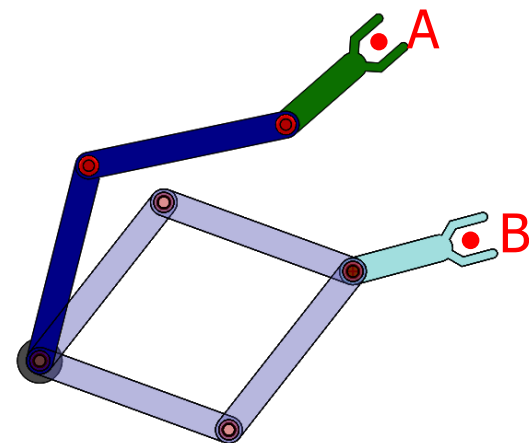
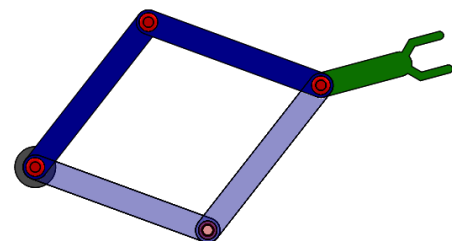
多重解 -3

□ 若具有多重解，解的選擇方式

◆ 離目前狀態最近的解

- 最快
- 最省能
-

◆ 避開障礙物





求解方法

- 解析法 Closed-form solutions
 - ◆ 用 代數algebraic 或 幾何geometric 方法
- 數值法 Numerical solutions
- 目前大多機械手臂設計成具有解析解
 - ◆ Pieper's solution: 相鄰三軸相交一點

Example: A RRR Manipulator -1

平面RRR (三个旋转关节) 机械臂

□ Ik problem: given (x, y, ϕ) , $(\theta_1, \theta_2, \theta_3) = ?$ 已知末端位姿, 反求关节角度

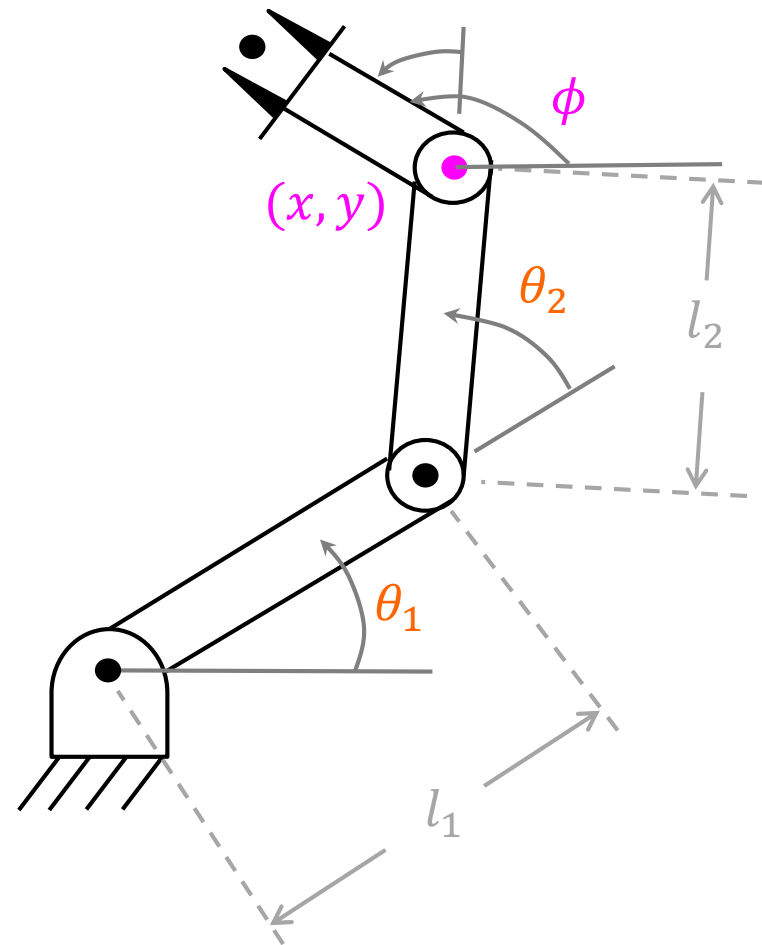
◆ Forward kinematics

$${}^0_3T = \begin{bmatrix} c_{123} & -s_{123} & 0.0 & l_1 c_1 + l_2 c_{12} \\ s_{123} & c_{123} & 0.0 & l_1 s_1 + l_2 s_{12} \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

◆ Goal point

$${}^0_3T = \begin{bmatrix} c_\phi & -s_\phi & 0.0 & x \\ s_\phi & c_\phi & 0.0 & y \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$C_{123} = \cos(\theta_1 + \theta_2 + \theta_3)$$



Example: A RRR Manipulator -2

- 幾何法：將空間幾何切割成平面幾何

$$x^2 + y^2 = l_1^2 + l_2^2 - 2l_1l_2\cos(180^\circ - \theta_2)$$

$$c_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1l_2}$$

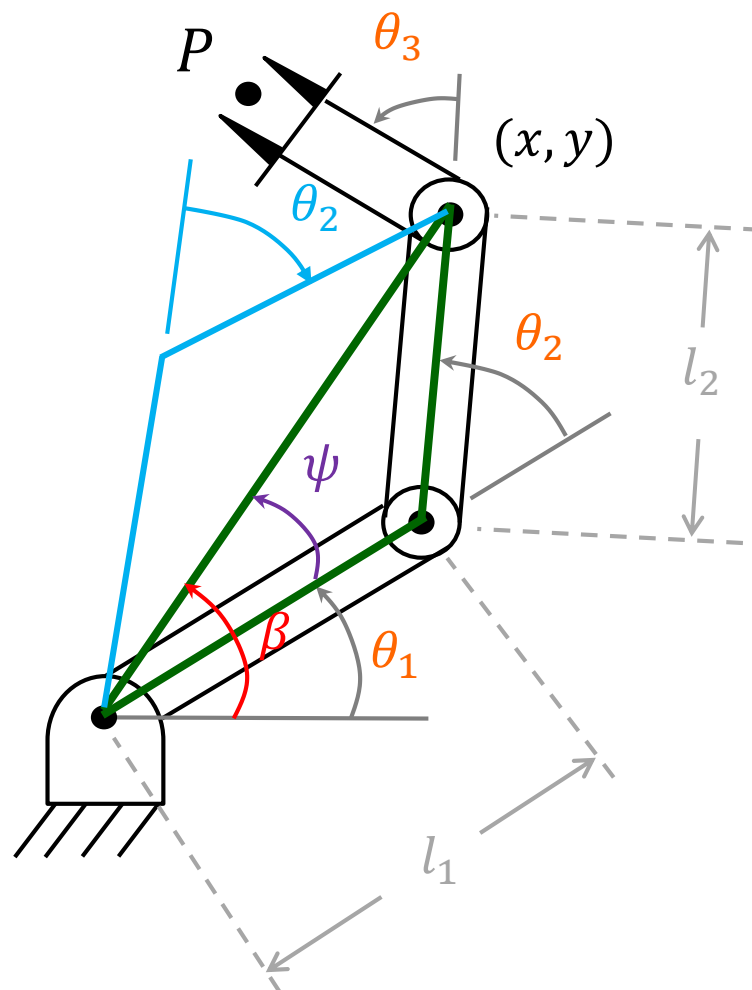
餘弦定理

$$\cos\psi = \frac{l_2^2 - (x^2 + y^2) - l_1^2}{-2l_1\sqrt{x^2 + y^2}}$$

三角形內角 $0^\circ < \psi < 180^\circ$

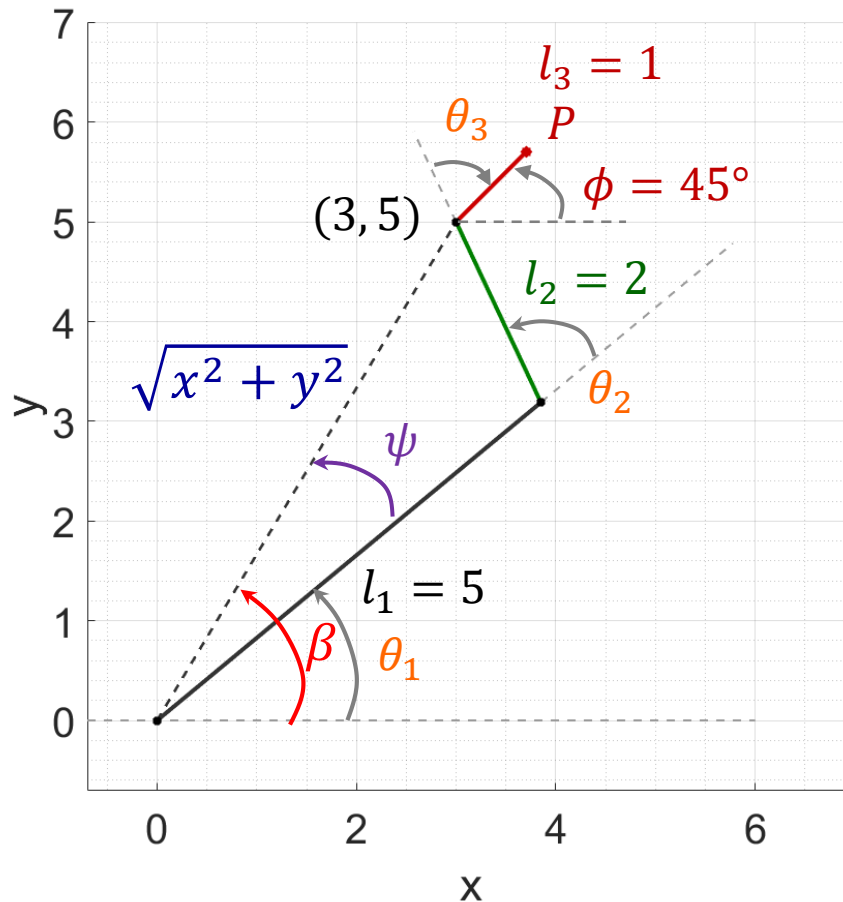
$$\theta_1 = \begin{cases} \text{atan2}(y, x) + \psi & \theta_2 < 0^\circ \\ \text{atan2}(y, x) - \psi & \theta_2 > 0^\circ \end{cases}$$

$$\theta_3 = \phi - \theta_1 - \theta_2$$



Example: A RRR Manipulator -3

□ Ex: 量化計算



$$c_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1l_2}$$

$$\theta_2 = 75.5^\circ$$

$$\cos\psi = \frac{l_2^2 - (x^2 + y^2) - l_1^2}{-2l_1\sqrt{x^2 + y^2}}$$

$$\psi = 19.4^\circ$$

$$\theta_1 = \text{atan2}(y, x) - \psi$$

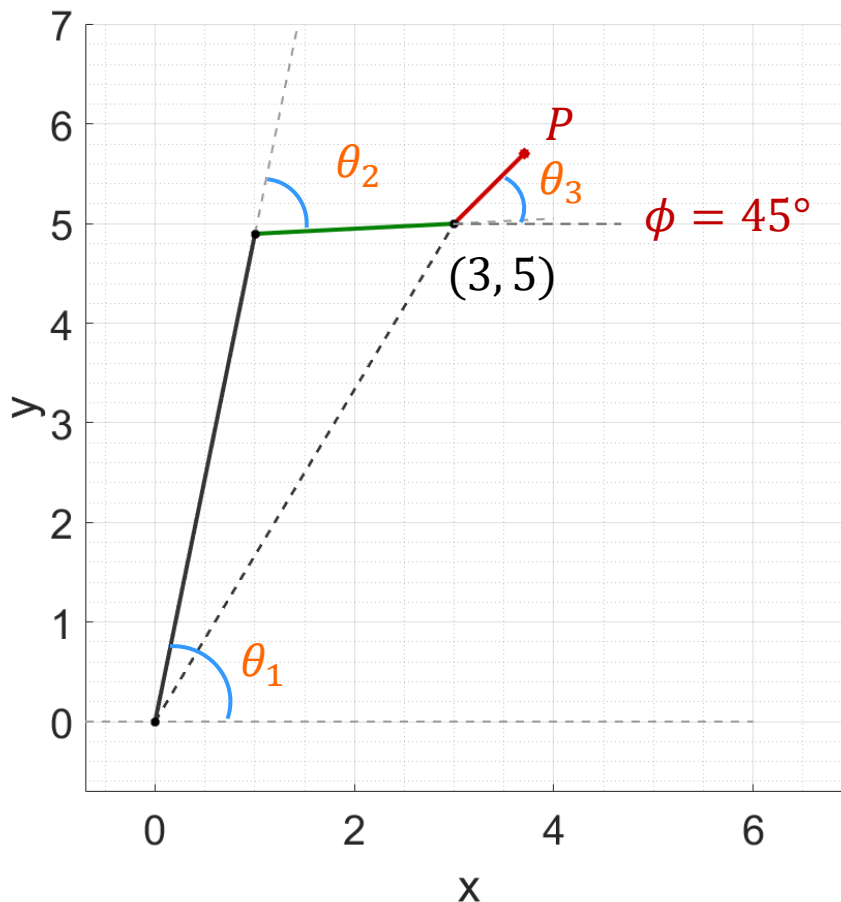
$$\theta_1 = 39.6^\circ$$

$$\theta_3 = \phi - \theta_1 - \theta_2$$

$$\theta_3 = -70.2^\circ$$

Example: A RRR Manipulator -4

- In-Video Quiz: 針對同一個位移和姿態，求得另一組 $(\theta_1, \theta_2, \theta_3)$ 的解



(A)

$$\theta_1 = 75.5$$

$$\theta_2 = -78.4$$

$$\theta_3 = 42.1$$

(B)

$$\theta_1 = 78.4$$

$$\theta_2 = -75.5$$

$$\theta_3 = 42.1$$

(C)

$$\theta_1 = -78.4$$

$$\theta_2 = 75.5$$

$$\theta_3 = 42.1$$

(D)

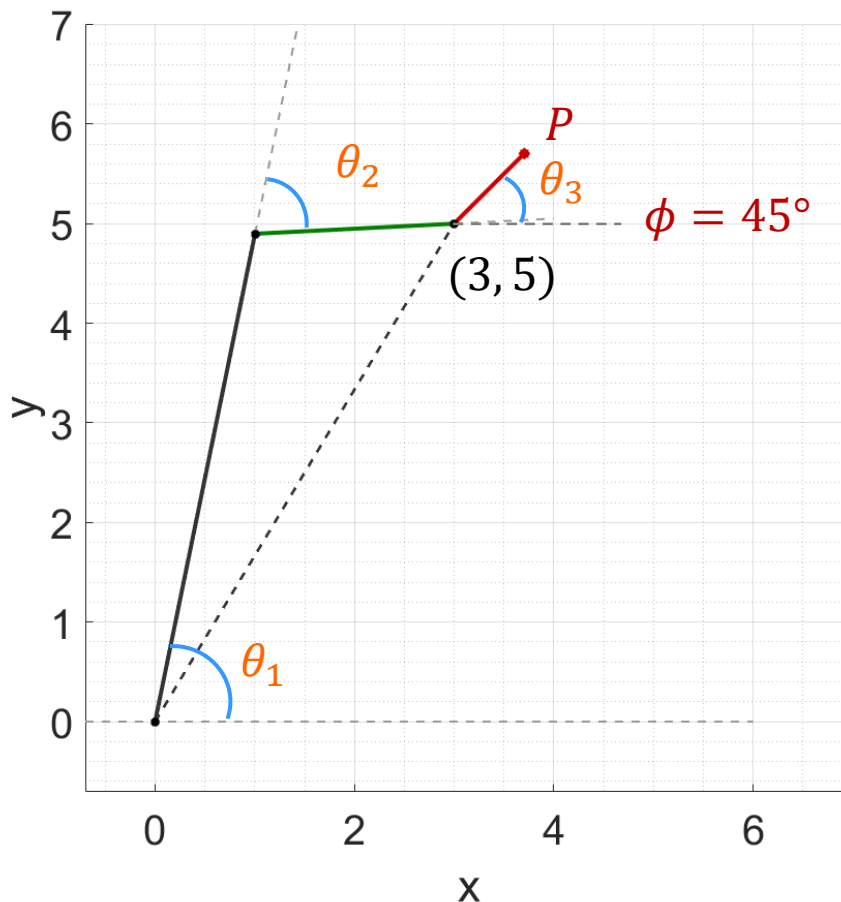
$$\theta_1 = 59$$

$$\theta_2 = -75.5$$

$$\theta_3 = 42.1$$

Example: A RRR Manipulator -4

- In-Video Quiz: 針對同一個位移和姿態，求得另一組 $(\theta_1, \theta_2, \theta_3)$ 的解



(A)
 $\theta_1 = 75.5$
 $\theta_2 = -78.4$
 $\theta_3 = 42.1$

(B)
 $\theta_1 = 78.4$
 $\theta_2 = -75.5$
 $\theta_3 = 42.1$

(C)
 $\theta_1 = -78.4$
 $\theta_2 = 75.5$
 $\theta_3 = 42.1$

(D)
 $\theta_1 = 59$
 $\theta_2 = -75.5$
 $\theta_3 = 42.1$

Example: A RRR Manipulator -5

□ 代數解

◆ 建立方程式

$$c_\phi = c_{123}$$

$$s_\phi = s_{123}$$

$$x = l_1 c_1 + l_2 c_{12}$$

$$y = l_1 s_1 + l_2 s_{12}$$

$$\begin{aligned} {}^0_3T &= \begin{bmatrix} c_{123} & -s_{123} & 0.0 & l_1 c_1 + l_2 c_{12} \\ s_{123} & c_{123} & 0.0 & l_1 s_1 + l_2 s_{12} \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} c_\phi & -s_\phi & 0.0 & x \\ s_\phi & c_\phi & 0.0 & y \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

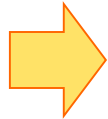
◆ 解 θ_2

$$x^2 + y^2 = l_1^2 + l_2^2 + 2l_1 l_2 c_2$$

$$c_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1 l_2}$$

> 1 or < -1 : too far for the manipulator to reach

$-1 \leq c_2 \leq 1$: "two solutions" $\theta_2 = \cos^{-1}(c_2)$



Example: A RRR Manipulator -6

- ◆ 將求得的 θ_2 帶入方程式

$$x = l_1 c_1 + l_2 c_{12} = (l_1 + l_2 c_2) c_1 + (-l_2 s_2) s_1 \triangleq k_1 c_1 - k_2 s_1$$

$$y = l_1 s_1 + l_2 s_{12} = (l_1 + l_2 c_2) s_1 + (l_2 s_2) c_1 \triangleq k_1 s_1 + k_2 c_1$$

- ◆ 變數變換

define

$$r = +\sqrt{k_1^2 + k_2^2}$$

then

$$k_1 = r \cos \gamma$$

$$\gamma = \text{Atan2}(k_2, k_1)$$

$$k_2 = r \sin \gamma$$

And then

$$\frac{x}{r} = \cos \gamma \cos \theta_1 - \sin \gamma \sin \theta_1 = \cos(\gamma + \theta_1)$$

$$\frac{y}{r} = \cos \gamma \sin \theta_1 + \sin \gamma \cos \theta_1 = \sin(\gamma + \theta_1)$$

Example: A RRR Manipulator -7

◆ 解 θ_1

$$\gamma + \theta_1 = \text{Atan2}\left(\frac{y}{r}, \frac{x}{r}\right) = \text{Atan2}(y, x)$$

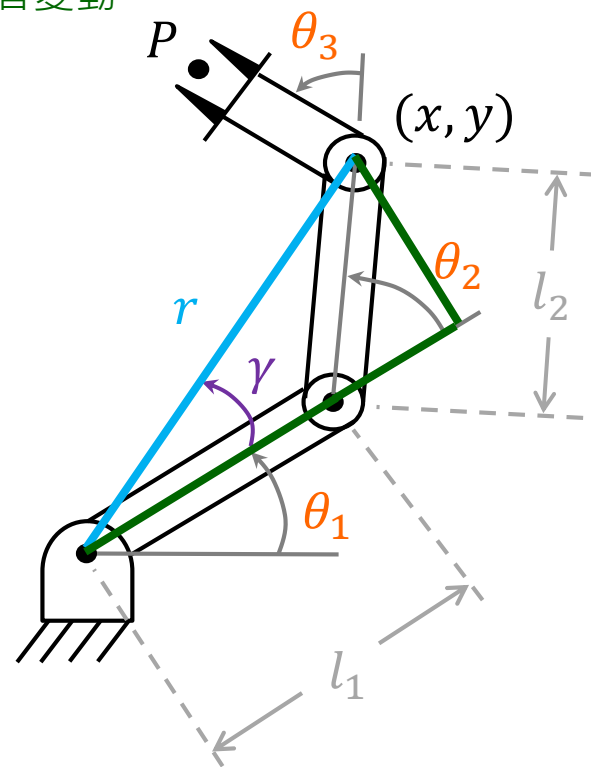
➡ $\theta_1 = \text{Atan2}(y, x) - \text{Atan2}(k_2, k_1)$

當 θ_2 選不同解， c_2 和 s_2 變動， k_1 和 k_2 變動， θ_1 也跟者變動

◆ 解 θ_3

$$\theta_1 + \theta_2 + \theta_3 = \text{Atan2}(s_\phi, c_\phi) = \phi$$

➡ $\theta_3 = \phi - \theta_1 - \theta_2$



三角函數方程式求解

□ Ex: 如何求得 $a\cos\theta + b\sin\theta = c$ 的 θ ?

◆ 方法：變換到polynomials (4階以下有解析解)

$$\tan\left(\frac{\theta}{2}\right) = u, \quad \cos\theta = \frac{1-u^2}{1+u^2}, \quad \sin\theta = \frac{2u}{1+u^2}$$

◆ 步驟：

$$a\cos\theta + b\sin\theta = c$$

$$a \frac{1-u^2}{1+u^2} + b \frac{2u}{1+u^2} = c$$

$$(a+c)u^2 - 2bu + (c-a) = 0$$

$$u = \frac{b \pm \sqrt{b^2 + a^2 - c^2}}{a+c}$$

a, b, c大小有限制, 不一定有解

$$\theta = 2 \tan^{-1}\left(\frac{b \pm \sqrt{b^2 + a^2 - c^2}}{a+c}\right) \quad a+c \neq 0$$

$$\theta = 180^\circ \quad a+c = 0$$

Pieper's Solution -1

□ 若6-DOF manipulator具有三個連續的軸交在同一點，則手臂有解析解

□ 一般，會把後三軸如此設計

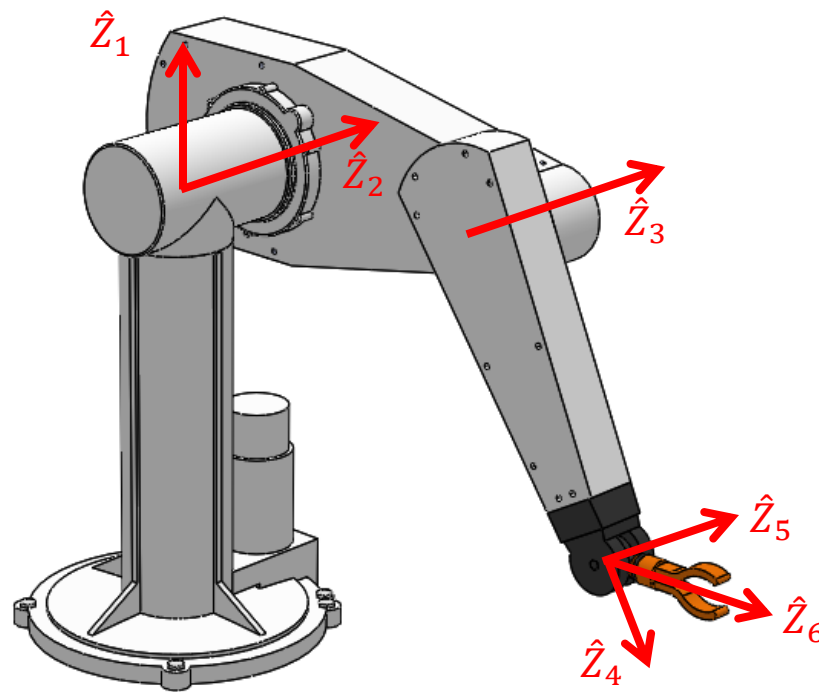
◆ 前三軸：產生移動

◆ 後三軸：產生轉動

□ Ex: A RRRRRR manipulator

◆ 因後三軸交一點

$${}^0P_{6\text{ ORG}} = {}^0P_{4\text{ ORG}}$$



Pieper's Solution -2

□ Positioning structure

◆ 法則：讓 $\theta_1, \theta_2, \theta_3$ 層層分離

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = {}^0P_4 \text{ ORG} = {}^0T_1 {}^1T_2 {}^2T_3 T^3 P_4 \text{ ORG}$$

$$\text{Note: } {}^{i-1}T_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\cos \theta_i = c\theta_i = c_i$$

$$\sin \theta_i = s\theta_i = s_i$$

$$= {}^0T_1 {}^1T_2 {}^2T_3 T \begin{bmatrix} a_3 \\ -d_4 s\alpha_3 \\ d_4 c\alpha_3 \\ 1 \end{bmatrix} = {}^0T_1 {}^1T_2 T \begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix}$$

so

4th column of 3T_4

$$\begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix} = {}^2T_3 \begin{bmatrix} a_3 \\ -d_4 s\alpha_3 \\ d_4 c\alpha_3 \\ 1 \end{bmatrix}$$

讓 $\theta_1, \theta_2, \theta_3$ 層層分離， f 為 θ_3 函數

$$f_1(\theta_3) = a_3 c_3 + d_4 s\alpha_3 s_3 + a_2$$

$$f_2(\theta_3) = a_3 c\alpha_2 s_3 - d_4 s\alpha_3 c\alpha_2 c_3 - d_4 s\alpha_2 c\alpha_3 - d_3 s\alpha_2$$

$$f_3(\theta_3) = a_3 s\alpha_2 s_3 - d_4 s\alpha_3 s\alpha_2 c_3 + d_4 c\alpha_2 c\alpha_3 + d_3 c\alpha_2$$

Pieper's Solution -3

◆ 下一步

$${}^0P_{4ORG} = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = {}^0T_1 {}^1T_2 T \begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix} = {}^0T_1 \begin{bmatrix} g_1(\theta_2, \theta_3) \\ g_2(\theta_2, \theta_3) \\ g_3(\theta_2, \theta_3) \\ 1 \end{bmatrix} = \begin{bmatrix} c_1 g_1 - s_1 g_2 \\ s_1 g_1 + c_1 g_2 \\ g_3 \\ 1 \end{bmatrix}$$

讓 $\theta_1, \theta_2, \theta_3$ 層層分離， g 為 θ_2, θ_3 函數

$$g_1(\theta_2, \theta_3) = c_2 f_1 - s_2 f_2 + a_1$$

$$g_2(\theta_2, \theta_3) = s_2 c \alpha_1 f_1 + c_2 c \alpha_1 f_2 - s \alpha_1 f_3 - d_2 s \alpha_1$$

$$g_3(\theta_2, \theta_3) = s_2 s \alpha_1 f_1 + c_2 s \alpha_1 f_2 + c \alpha_1 f_3 + d_2 c \alpha_1$$

$$r = x^2 + y^2 + z^2 = g_1^2 + g_2^2 + g_3^2 \quad r \text{ 僅為 } \theta_2, \theta_3 \text{ 函數}$$

$$= f_1^2 + f_2^2 + f_3^2 + a_1^2 + d_2^2 + 2d_2 f_3 + 2a_1(c_2 f_1 - s_2 f_2)$$

$$= (k_1 c_2 + k_2 s_2) 2a_1 + k_3$$

$$k_1(\theta_3) = f_1$$

$$k_2(\theta_3) = -f_2$$

$$k_3(\theta_3) = f_1^2 + f_2^2 + f_3^2 + a_1^2 + d_2^2 + 2d_2 f_3$$

Pieper's Solution -4

◆ 此外

$$z = g_3 = (k_1 s_2 - k_2 c_2) s \alpha_1 + k_4 \quad z \text{ 僅為 } \theta_2, \theta_3 \text{ 函數}$$

$$k_1(\theta_3) = f_1$$

$$k_2(\theta_3) = -f_2$$

$$k_4(\theta_3) = f_3 c \alpha_1 + d_2 c \alpha_1$$

◆ 整合 r 和 z 一起考量

$$\begin{cases} r = (k_1 c_2 + k_2 s_2) 2a_1 + k_3 \\ z = (k_1 s_2 - k_2 c_2) s \alpha_1 + k_4 \end{cases}$$

◦ If $a_1 = 0$, $r = k_3(\theta_3) = f_1^2 + f_2^2 + f_3^2 + a_1^2 + d_2^2 + 2d_2 f_3$

◦ If $s \alpha_1 = 0$, $z = k_4(\theta_3) = f_3 c \alpha_1 + d_2 c \alpha_1$

◦ Else

$$\frac{(r - k_3)^2}{4a_1^2} + \frac{(z - k_4)^2}{s^2 \alpha_1} = k_1^2 + k_2^2$$



Solve θ_3 of all three cases by using " $u = \tan\left(\frac{\theta_3}{2}\right)$ "

Pieper's Solution -5

□ 最後

Using $r = (k_1 c_2 + k_2 s_2) 2a_1 + k_3$ to solve θ_2

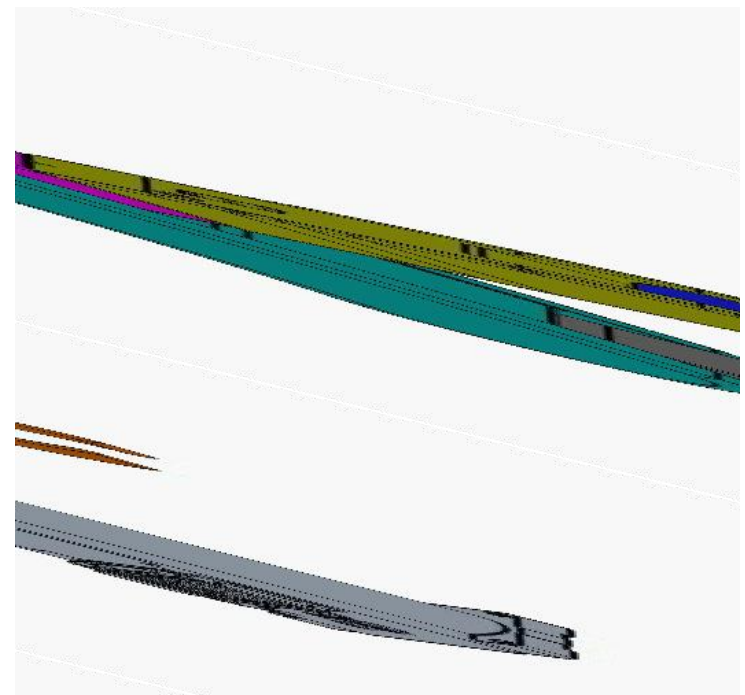
Using $x = c_1 g_1(\theta_2, \theta_3) - s_1 g_2(\theta_2, \theta_3)$ to solve θ_1

□ Orientation

◆ $\theta_1, \theta_2, \theta_3$ 已知

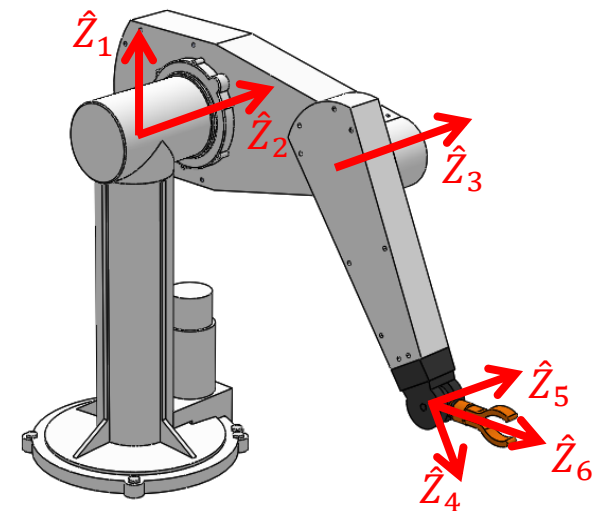
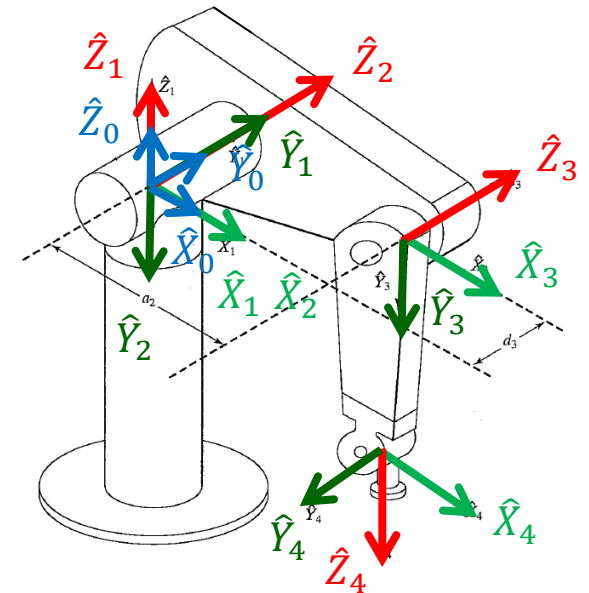
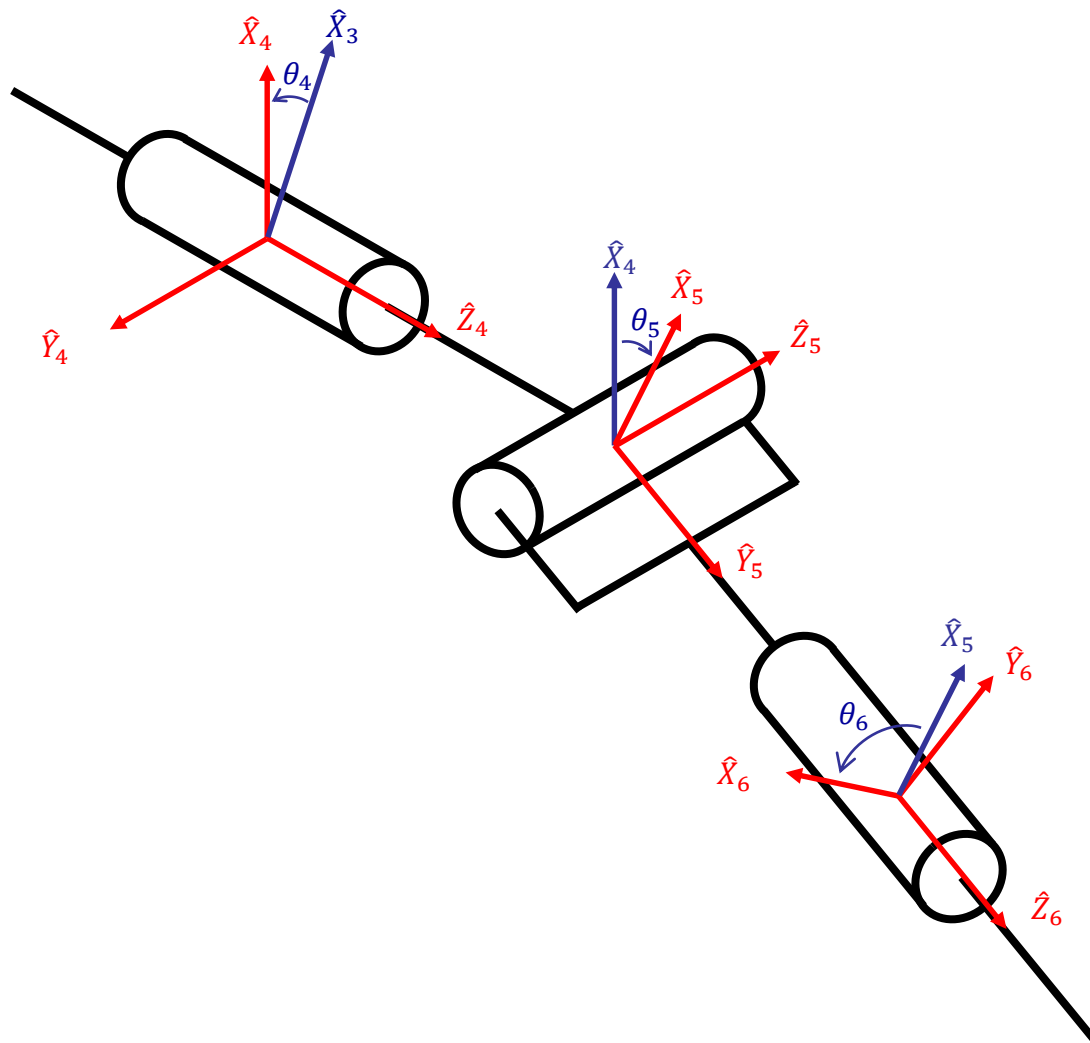
$${}^3_6R = {}^0_3R^{-1} {}^0_6R$$

◆ 以 Z-Y-Z Euler angle 求解 $\theta_4, \theta_5, \theta_6$



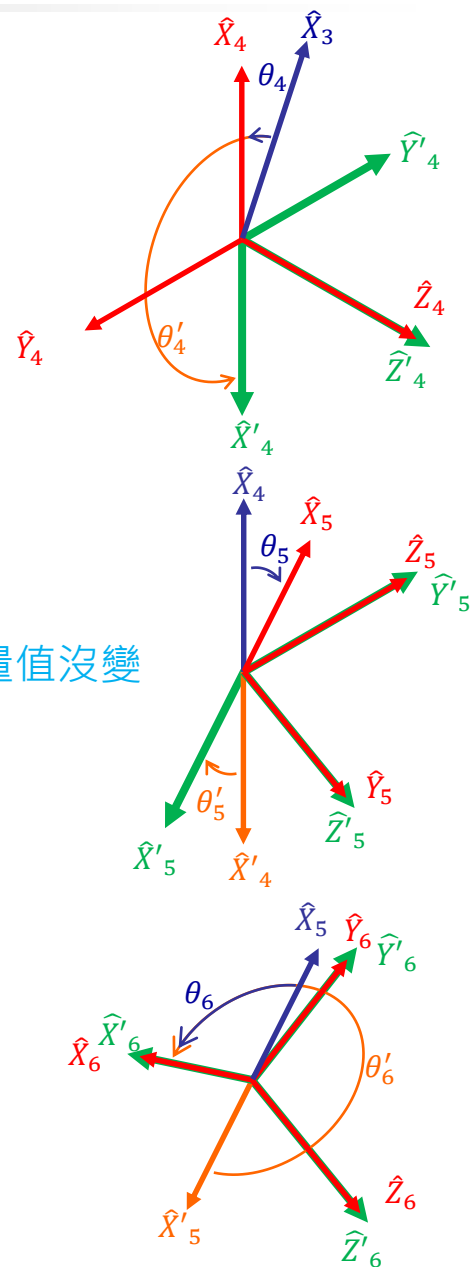
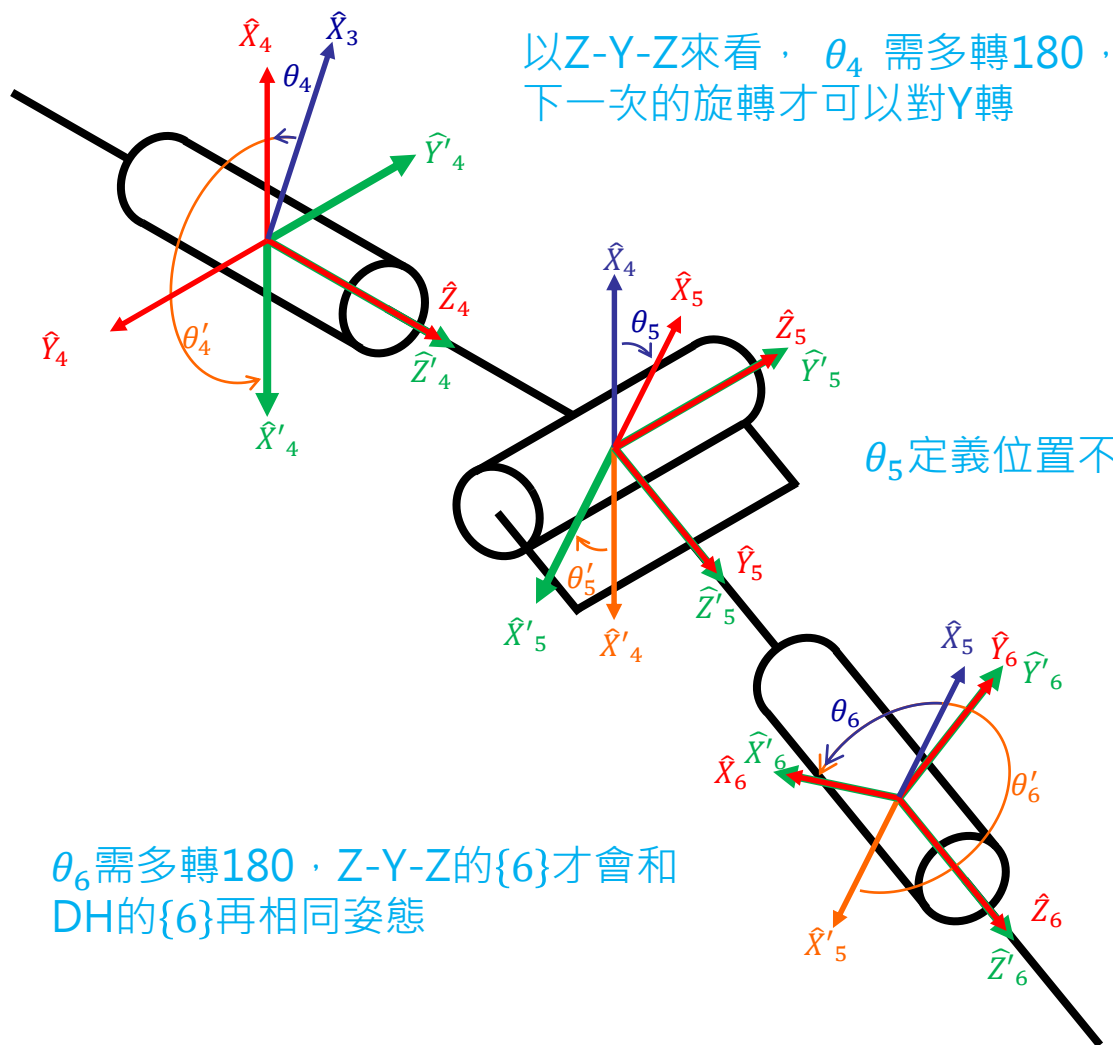
Pieper's Solution -6

- Joints 4-6, DH definition



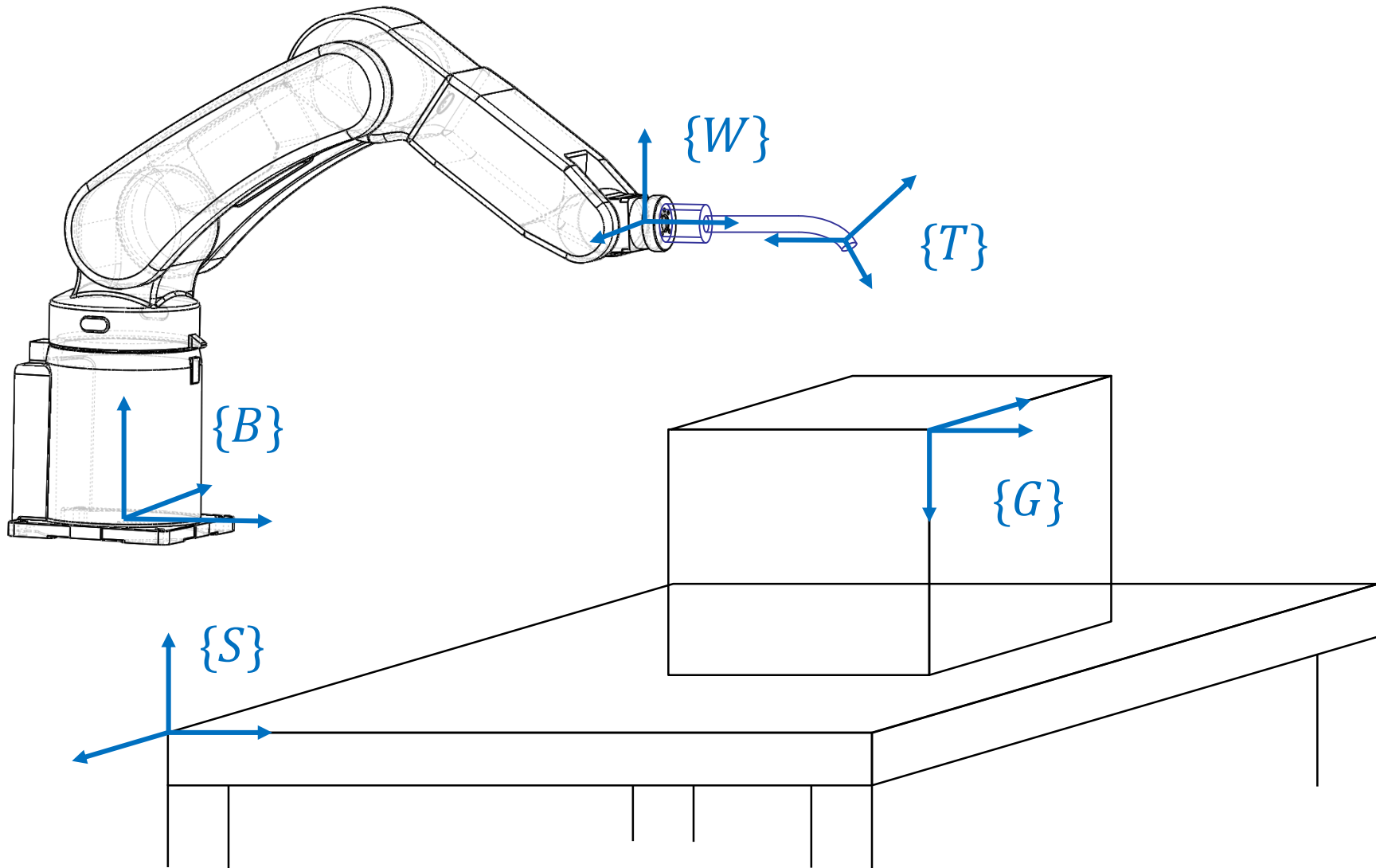
Pieper's Solution -7

□ DH definition vs. Z-Y-Z Euler Angles



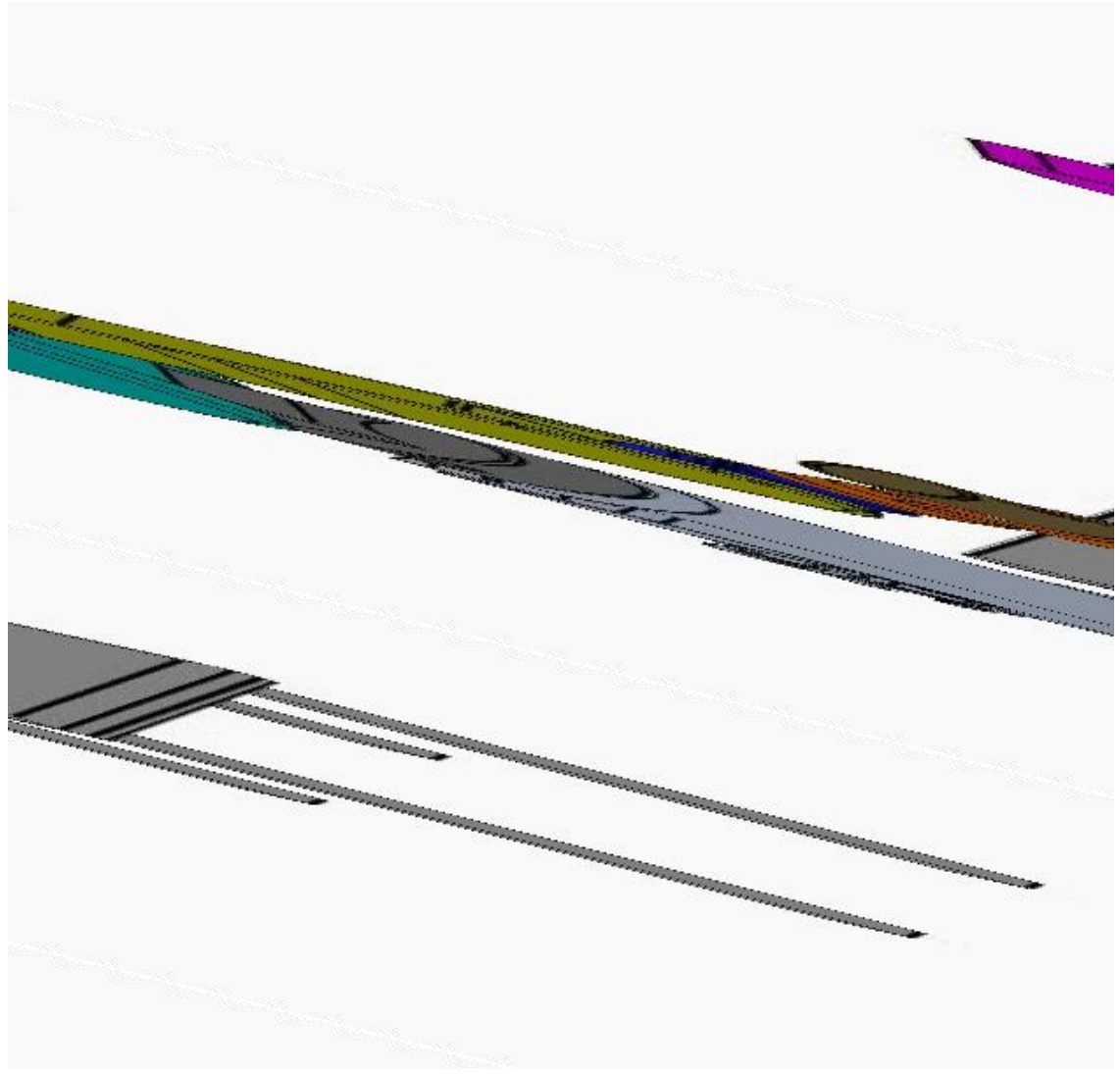
座標系

- Base, wrist, tool, station, and goal frames



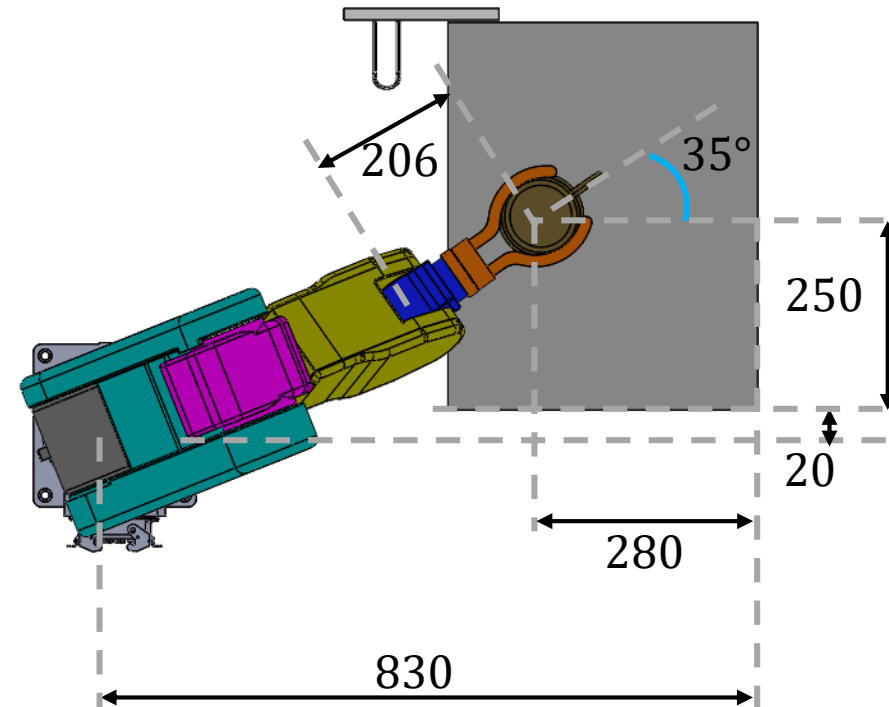
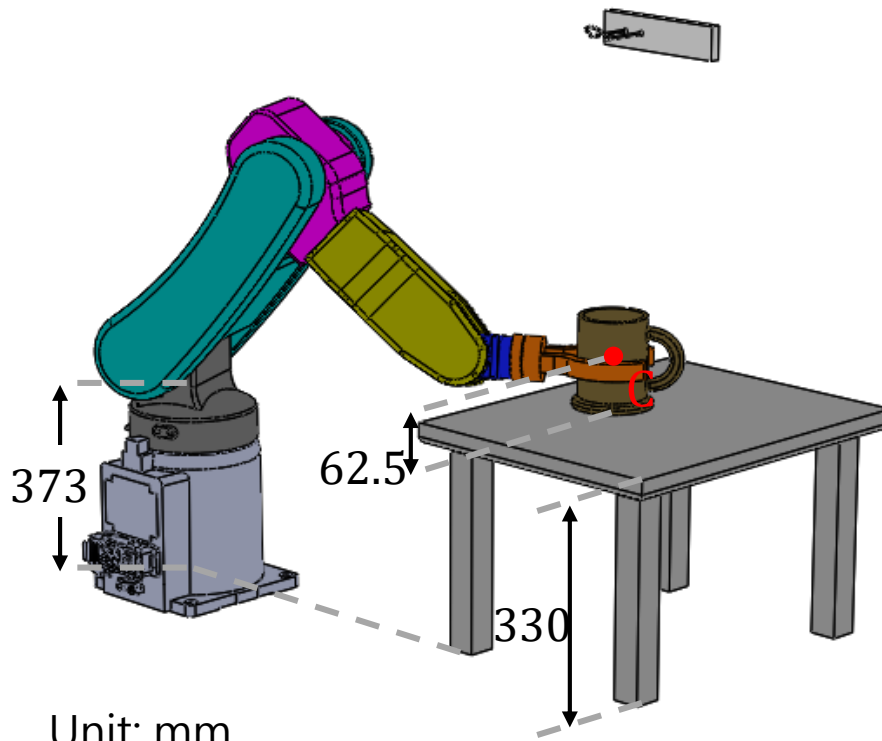
Example: 物件取放任務 -1

- 情境：機械手臂夾住放在桌上的杯子，移動手臂將杯子掛到牆上的杯架



Example: 物件取放任務 -2

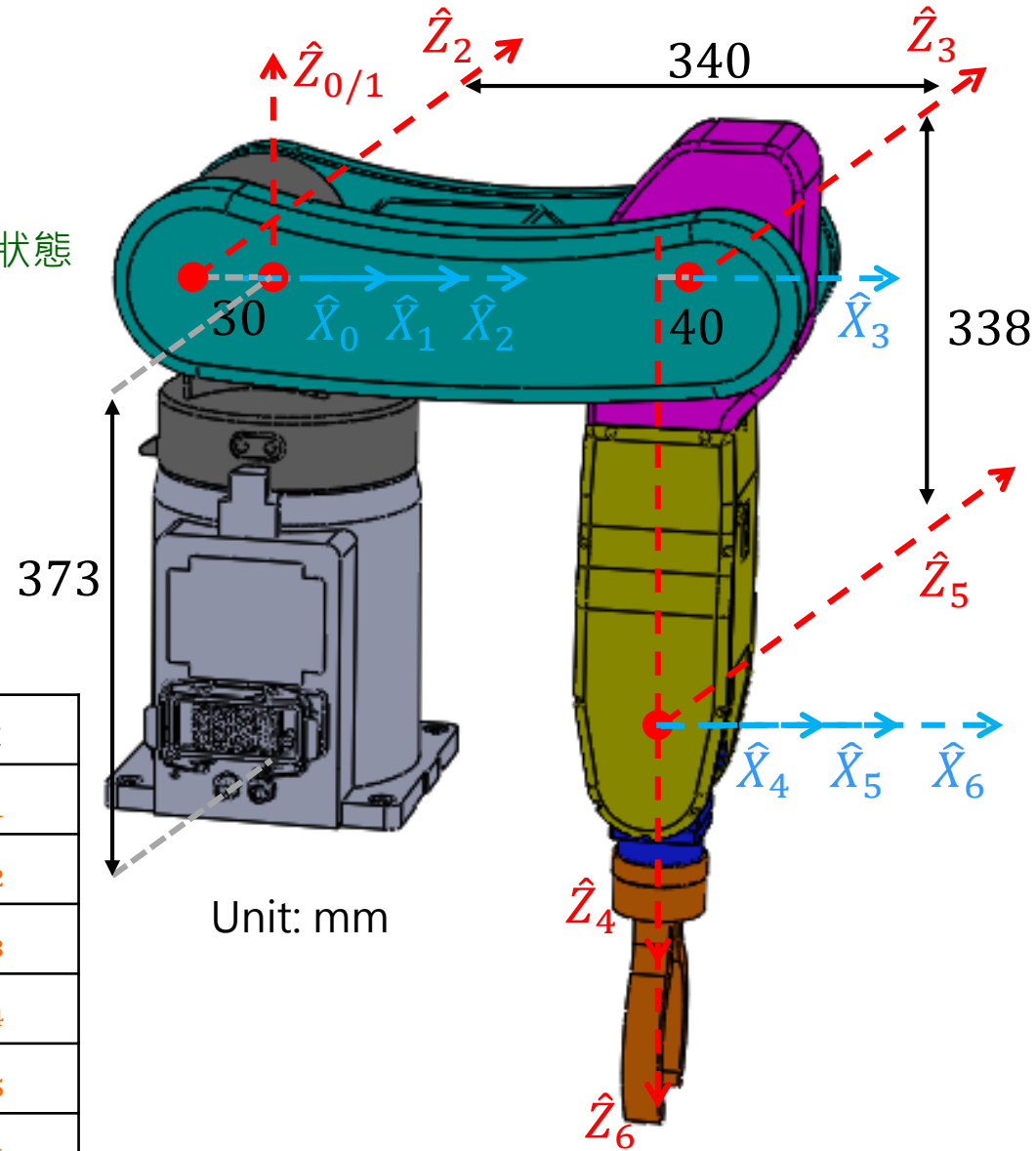
- 現階段任務：為使RRRRRR手臂能以下圖姿態夾住杯子（任務的起始點C），手臂的6個joint angles需為何？



Example: 物件取放任務 -3

□ Step 1: 定義DH Table

圖中顯示各軸為0°的狀態



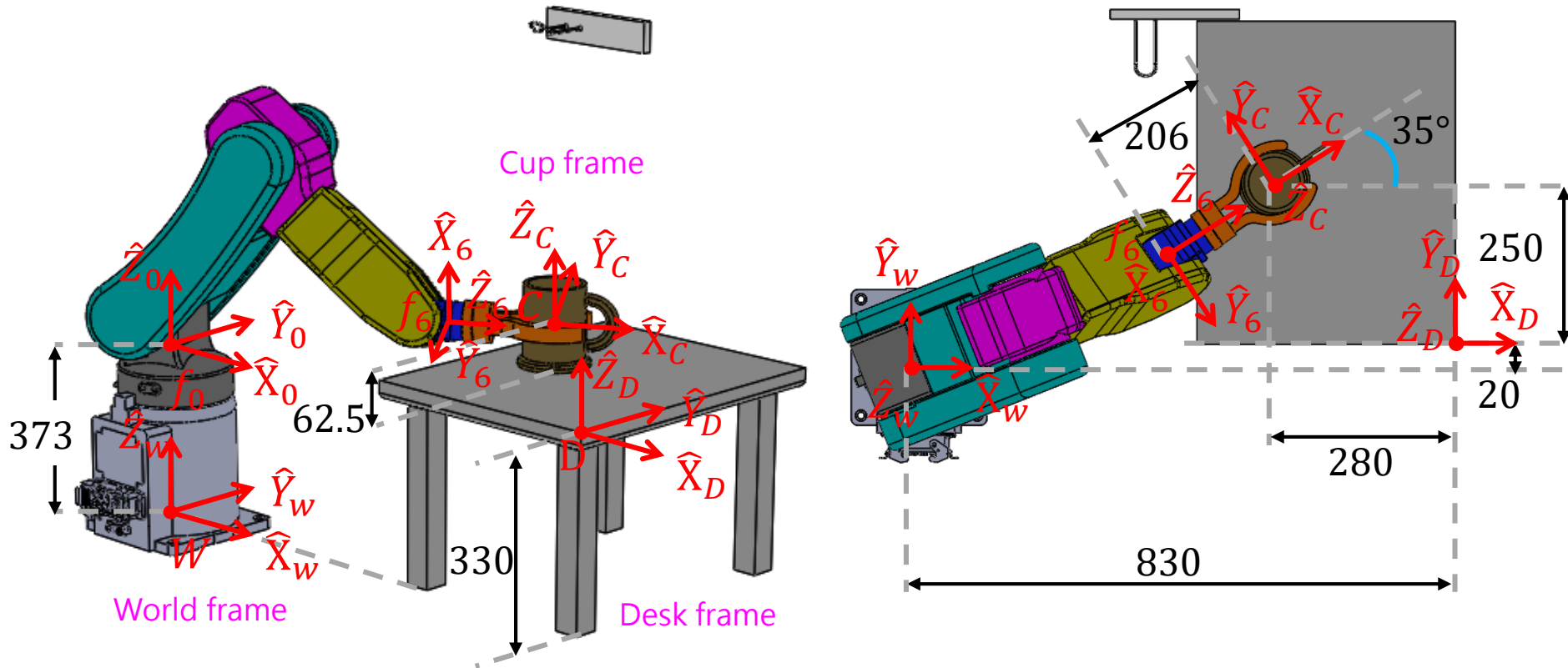
i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0°	0	0	θ_1
2	-90°	$a_1 = -30$	0	θ_2
3	0°	$a_2 = 340$	0	θ_3
4	-90°	$a_3 = -40$	$d_4 = 338$	θ_4
5	90°	0	0	θ_5
6	-90°	0	0	θ_6

Example: 物件取放任務 -4

□ Step 2: 找出 ${}^W_C T$ ，再進一步找出 ${}^0_6 T$

$${}^W_C T = {}^W_D T {}^D_C T = \begin{bmatrix} 1 & 0 & 0 & 830 \\ 0 & 1 & 0 & 20 \\ 0 & 0 & 1 & 330 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 35^\circ & -\sin 35^\circ & 0 & -280 \\ \sin 35^\circ & \cos 35^\circ & 0 & 250 \\ 0 & 0 & 1 & 62.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

由「桌子相對於手臂」和「杯子相對於桌子」的相對關係推得



Example: 物件取放任務 -5

$${}^W_c T = {}^W_0 T {}^0_6 T {}^6_c T$$

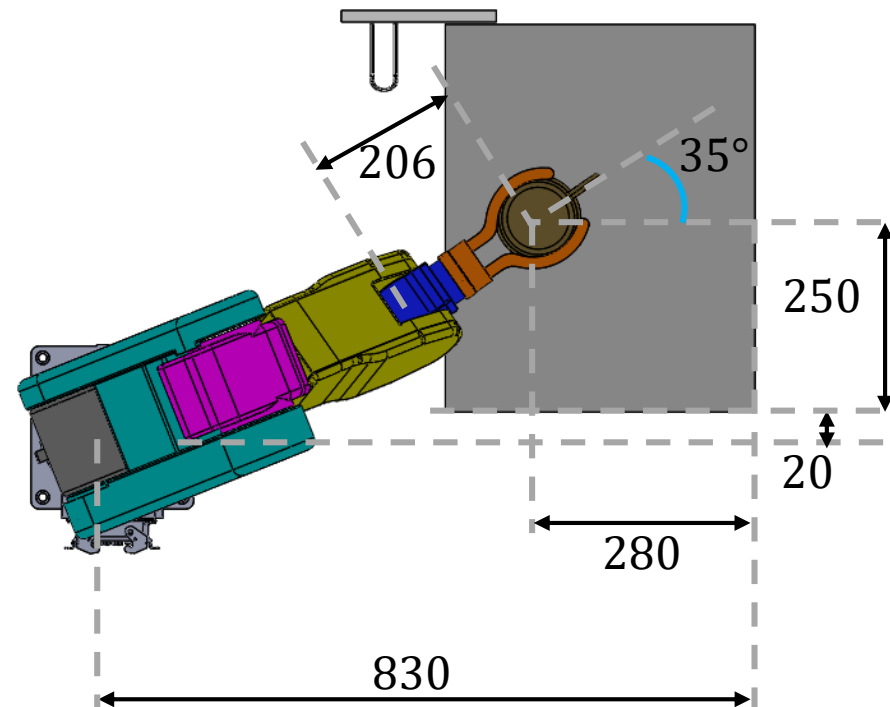
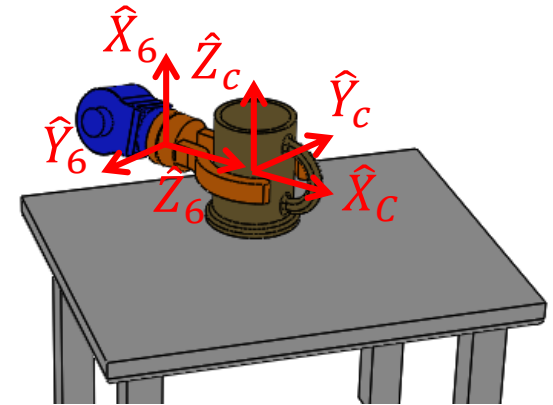
$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 373 \\ 0 & 0 & 0 & 1 \end{bmatrix} {}^0_6 T \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 206 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_6 T = {}^W_0 T^{-1} {}^W_c T {}^6_c T^{-1}$$

$$= \begin{bmatrix} 0 & 0.5736 & 0.8192 & 381.3 \\ 0 & -0.8192 & 0.5736 & 151.8 \\ 1 & 0 & 0 & 19.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_6 R = \begin{bmatrix} 0 & 0.5736 & 0.8192 \\ 0 & -0.8192 & 0.5736 \\ 1 & 0 & 0 \end{bmatrix}$$

$${}^0 P_{6\ ORG} = \begin{bmatrix} 381.3 \\ 151.8 \\ 19.5 \end{bmatrix}$$



Example: 物件取放任務 -6

□ Step 3: 找出 $\theta_1 - \theta_6$

◆ $\theta_1 \theta_2 \theta_3$ 角度求解

$$\begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix} = {}^2_3T {}^3P_{4ORG}$$
$$= \begin{bmatrix} c_3 & -s_3 & 0 & 340 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -40 \\ 338 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 340 - 338s_3 - 40c_3 \\ 338c_3 - 40s_3 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} g_1(\theta_2, \theta_3) \\ g_2(\theta_2, \theta_3) \\ g_3(\theta_2, \theta_3) \\ 1 \end{bmatrix} = {}^1_2T \begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix}$$
$$= \begin{bmatrix} c_2 & -s_2 & 0 & -30 \\ 0 & 0 & 1 & 0 \\ -s_2 & -c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix} = \begin{bmatrix} 340c_2 - 40c_{23} - 338s_{23} - 30 \\ 0 \\ 40s_{23} - 338c_{23} - 340s_2 \\ 1 \end{bmatrix}$$

Example: 物件取放任務 -7

$$r = (k_1 c_2 + k_2 s_2) 2a_1 + k_3 = ||P||^2 = 168813.18$$

$$z = (k_1 s_2 - k_2 c_2) s \alpha_1 + k_4 = 19.5$$

計算 θ_1 θ_2 θ_3 角度

$$\frac{(r-k_3)^2}{4a_1^2} + \frac{(z-k_4)^2}{s^2 \alpha_1} = k_1^2 + k_2^2$$

$$\Rightarrow \text{solve } \theta_3 = 2.5^\circ$$

$$r = (k_1 c_2 + k_2 s_2) 2a_1 + k_3$$

$$\Rightarrow \text{solve } \theta_2 = -52.2^\circ$$

$$x = c_1 g_1(\theta_2, \theta_3) - s_1 g_2(\theta_2, \theta_3)$$

$$\Rightarrow \text{solve } \theta_1 = 21.8^\circ$$

Example: 物件取放任務 -8

◆ θ_4 θ_5 θ_6 角度求解

$${}^0_3R = \begin{bmatrix} 0.6006 & 0.7082 & -0.3710 \\ 0.24 & 0.2830 & 0.9286 \\ 0.7627 & -0.6468 & 0 \end{bmatrix}$$

$${}^3_6R = {}^0_3R^{-1} {}^0_6R = \begin{bmatrix} 0.7627 & 0.1477 & 0.6297 \\ -0.6468 & 0.1744 & 0.7424 \\ 0 & -0.9735 & 0.2286 \end{bmatrix}$$

使用 Z-Y-Z Euler angle 求得剩下的 joint angles

$$\theta_4 = -20^\circ \quad \theta_5 = -42^\circ \quad \theta_6 = 15^\circ$$

Example: 物件取放任務 -8

- Class Exercise: 要讓夾爪到杯子的末端點E，6個轉軸的轉角又分別是？

